Ilia Karmanov

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UK and EU citizen (no visa sponsorship needed)

Overview

Senior Staff Research Scientist at NVIDIA ADLR, working on multimodal architecture, pre-training, post-training, and evaluation for NVIDIA's Nemotron models. First author of Eclair (shipped open-source as Nemotron Parse), a document-understanding model that generated 50B+ tokens for LLM pre-training and underpins NVIDIA's VLM document understanding. Earlier, published at NeurIPS, ICCV, and BMVC at Qualcomm AI Research. Came to machine learning in 2016 from economics: an MSc at LSE focused on optimal control and causal inference, and development-economics research at the International Growth Centre (Oxford and LSE).

Experience

NVIDIA ADLR, Zurich

Senior Staff Research Scientist *Mar 2025 – present*

- Work on post-training for Nemotron multimodal models, using synthetic data, SFT, RL and distillation to improve existing capabilities and build new ones. Start from evaluation failures, work out why the model is failing, and design and test the intervention needed to fix it
- Design training curricula and adapt algorithms to the task, including how methods are sequenced and how objectives, rewards, sampling and optimisation are defined
- Made the case for RL post-training and drove the first GRPO proof of concept with the systems team; it improved the model and became the starting point for the team's standard RL pipeline
- Current research focuses on multi-teacher distillation for smaller models and adapting RL to multimodal and long-context tasks

Staff Research Scientist *Sep 2022 – Feb 2025*

- Led Eclair from architecture and data through training and evaluation, and was first author of the resulting paper. Released as the open-source Nemotron Parse, it was adopted across NVIDIA to generate 50B+ pre-training tokens and distil VLMs; it also underpinned Llama Nemotron Nano VL, which topped OCRBench v2 on release
- Designed its asymmetric architecture: a larger vision encoder with a lightweight decoder, no decoder positional embeddings, multi-token prediction, and one prompt-controlled output format spanning text, layout and semantic classes
- Contributed more broadly to multimodal pre-training and architecture development for Nemotron LLMs and VLMs

Qualcomm AI Research, Amsterdam

Staff Research Scientist *Dec 2021 – Jul 2022*

- Developed hardware-efficient architectures for 4K video-to-video processing (denoising on device)
- Efficient Mixture-of-Experts architecture for conditional computation (BMVC 2022)

Senior Research Scientist *Feb 2020 – Nov 2021*

- Created the first weakly-supervised approach for 3D person-localisation on video and WiFi data, published at NeurIPS 2021 (with Max Welling) and IEEE GLOBECOM 2021 (first author)
- Video recognition via motion-augmented self-training (ICCV 2021)

Microsoft, London

Senior Applied Scientist *Dec 2018 – Jan 2020*

- Applied computer vision across industry problems: self- and fully-supervised segmentation (medical and seismic imaging), object detection and counting, and real-time video action recognition
- Built synthetic-data pipelines to improve model generalisation

Applied Scientist *Jun 2016 – Nov 2018*

- Collaborated with MSR on Z3 theorem-prover for plane-gate allocations (Mixed Integer Linear Programming)
- Created DeepLearningFrameworks, a cross-framework benchmark that drew 1,700+ GitHub stars and advice from framework authors including Soumith Chintala (PyTorch) and Yangqing Jia (Caffe)

Charles River Associates, London
Consultant *May 2014 – Jun 2016*

- Econometrics for antitrust and M&A cases: regression specification and robustness, mathematical programming, and geospatial algorithms
- Initiated the first data-scientist role within the department

International Growth Centre, University of Oxford
Research Economist *Oct 2013 – May 2014*

- Applied econometrics for development economics at the IGC, the Oxford and LSE research centre funded by the UK government, under the firm-capabilities theme (large firms, entrepreneurship, trade) directed by Prof. Paul Collier
- Performed fieldwork in India, Pakistan, and Uganda with local researchers, presenting findings to policymakers

Publications

I Karmanov, A Deshmukh, L Voegtle, P Fischer, K Chumachenko, T Roman, J Seppanen, J Parmar, J Jennings, A Tao, K Sapra. Eclair: Extracting Content and Layout with Integrated Reading Order for Documents. arXiv 2025

K Chumachenko, A Deshmukh, J Seppanen, **I Karmanov**, C Chen, L Voegtle, P Fischer, et al. Nemotron Parse 1.1. arXiv 2025

Contributing author on NVIDIA tech reports: Nemotron 3 Nano Omni (arXiv 2026), Nemotron Nano V2 VL (arXiv 2025), and Nemotron-H Hybrid Mamba-Transformer (arXiv 2025)

Z Li, G Chen, S Liu, ... **I Karmanov**, L Voegtle, P Fischer, ... Z Yu. Eagle 2: Building Post-Training Data Strategies from Scratch for Frontier Vision-Language Models. NeurIPS 2025

A Royer, **I Karmanov**, A Skliar, B Ehteshami Bejnordi, T Blankevoort. Revisiting Single-gated Mixtures of Experts. BMVC 2022

FG Zanjani, **I Karmanov**, H Ackermann, D Dijkman, S Merlin, M Welling, F Porikli. Modality-Agnostic Topology Aware Localization. NeurIPS 2021

I Karmanov, FG Zanjani, S Merlin, I Kadampot, D Dijkman. WiCluster: Passive Indoor 2D/3D Positioning using WiFi without Precise Labels. IEEE GLOBECOM 2021

K Gavrilyuk, M Jain, **I Karmanov**, CGM Snoek. Motion-Augmented Self-Training for Video Recognition at Smaller Scale. ICCV 2021

Y Ren, J Lu, A Beletchi, Y Huang, **I Karmanov**, et al. Hand Gesture Recognition using 802.11ad mmWave. IEEE WCNC 2021

Book: M Salvaris, D Dean, WH Tok, **I Karmanov**. Deep Learning with Azure. Apress, 2018

J Costa-Font, F Cowell (with research contributions from **I Karmanov**). European Identity and Redistributive Preferences. CESifo Working Paper, 2015

Patents

12 US patent applications filed at Qualcomm (several granted), across RF positioning, gesture recognition, and efficient architectures.

Service

Reviewer for NeurIPS, ICLR, and ICML.

Education

London School of Economics
MSc Economics *2012 – 2013*

- Thesis: Optimal Control Analysis for Tax Avoidance. Modelling a firm's optimisation problem in an infinite horizon setting, introducing reputation as an accumulated asset, solved with Hamiltonian in an optimal-control setting. Advisor: Prof. Frank Cowell

BSc Economics, First Class Honours *2009 – 2012*